

WARWICK, Australia

WMO: 945550

Lat: 28.2061S

Lon: 152.1003E

Elev: 1561

StdP: 13.89

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	30.1	33.1	18.6	14.8	55.7	23.2	18.5	57.4	19.2	58.3	17.0	59.2	0.7	60	0.406

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	20.9	93.1	68.5	90.1	68.0	87.2	67.2	73.7	84.9	71.9	82.4	70.7	80.7	10.0	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.8	120.4	77.6	69.0	112.9	75.9	67.6	107.6	74.8	38.4	85.0	36.8	82.4	35.6	80.9	95.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.1	15.8	14.1	DB	24.5	99.1	2.4	3.2	22.7	101.4	21.3	103.3	19.9	105.1	18.2	107.4	(4)
(5)			WB	27.5	79.5	3.2	7.1	25.2	84.7	23.4	88.8	21.6	92.9	19.3	98.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	63.5	74.2	73.1	70.4	64.3	57.2	52.9	51.2	53.0	59.4	64.7	69.4	72.4	(6)	
	DBStd	9.34	4.10	4.29	3.47	3.94	4.60	4.76	4.42	4.74	4.99	4.98	5.14	4.78	(7)	
	HDD50	91	0	0	0	0	4	26	39	21	1	0	0	0	(8)	
	HDD65	1732	0	1	2	57	243	363	429	373	179	66	16	3	(9)	
	CDD50	5002	749	647	632	429	229	113	75	114	283	456	581	694	(10)	
	CDD65	1168	285	228	168	36	2	0	0	2	12	57	146	232	(11)	
	CDH74	10225	2446	1825	1214	307	26	1	1	45	281	711	1382	1986	(12)	
	CDH80	3491	948	665	324	36	2	0	0	11	59	198	498	750	(13)	
(14) Wind	WSAvg	6.0	7.2	6.7	6.1	5.4	4.7	5.1	5.0	5.5	6.1	6.4	7.1	7.1	(14)	
(15) Precipitation	PrecAvg	28.7	3.3	3.3	2.5	1.5	2.1	1.1	1.7	1.3	1.3	2.8	3.1	4.4	(15)	
	PrecMax	43.5	7.6	10.5	7.9	12.1	9.8	3.9	5.7	3.6	4.1	7.7	7.2	11.2	(16)	
	PrecMin	17.6	0.2	0.3	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.6	0.6	0.9	(17)	
	PrecStd	6.1	1.9	2.2	2.1	2.1	1.9	0.9	1.5	0.9	1.0	1.7	1.8	2.4	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	96.1	96.6	92.0	83.9	77.2	71.2	71.5	79.3	85.9	90.0	94.1	96.1	(19)	
		MCWB	71.2	70.4	68.2	63.6	60.9	58.5	55.6	57.6	60.5	62.8	66.0	67.2	(20)	
	2%	DB	92.1	90.9	86.7	80.3	73.3	68.2	67.8	73.1	81.0	85.1	89.8	91.7	(21)	
		MCWB	69.3	70.7	67.7	63.0	59.0	56.8	53.9	55.0	58.8	61.9	64.8	67.8	(22)	
	5%	DB	88.8	87.2	83.5	77.5	70.9	65.7	65.2	69.4	77.0	81.4	85.7	87.7	(23)	
		MCWB	69.1	69.3	66.9	62.5	57.8	55.3	53.1	53.5	57.7	61.5	64.4	67.3	(24)	
	10%	DB	85.3	83.9	80.7	74.9	68.3	63.2	62.9	66.2	73.1	77.7	81.9	84.2	(25)	
		MCWB	68.5	68.7	66.0	61.8	57.0	54.1	52.2	52.3	56.7	60.6	63.9	67.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	79.7	77.3	72.7	69.3	65.8	62.0	61.9	61.1	64.6	69.8	71.1	77.0	(27)	
		MCDB	85.7	86.6	82.9	77.7	69.8	66.9	66.5	72.3	76.4	79.2	85.0	79.1	(28)	
	2%	WB	73.6	73.6	70.7	66.5	62.6	59.4	58.5	58.3	62.3	66.3	69.1	71.7	(29)	
		MCDB	84.9	84.7	79.8	74.6	69.7	64.6	63.3	67.4	73.5	77.3	81.5	83.1	(30)	
	5%	WB	71.9	71.8	69.2	64.9	60.6	57.6	56.0	56.2	60.5	64.4	67.5	70.2	(31)	
		MCDB	82.7	81.4	78.2	73.2	66.9	62.6	61.6	65.1	71.1	75.2	79.1	80.8	(32)	
	10%	WB	70.4	70.4	67.8	63.4	59.0	55.9	54.0	54.4	58.9	62.7	66.0	68.9	(33)	
		MCDB	80.4	79.6	76.3	71.6	65.4	61.1	60.1	63.5	69.5	72.9	76.4	78.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	20.9	20.2	20.7	21.9	23.2	21.4	24.3	27.6	27.0	24.4	22.6	21.3	(35)	
		MCDBR	27.5	26.7	26.4	26.6	27.6	25.5	29.2	33.8	34.3	31.6	29.2	27.8	(36)	
	5% WB	MCWBR	8.7	8.7	9.4	11.2	13.9	14.0	15.3	16.7	14.6	12.0	9.7	9.0	(37)	
		MCWBR	21.6	21.5	20.1	21.2	19.9	18.7	20.2	25.3	25.3	24.2	22.7	21.2	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.361	0.350	0.331	0.313	0.292	0.287	0.279	0.292	0.330	0.350	0.363	0.367	(40)	
		taud	2.529	2.574	2.624	2.645	2.650	2.666	2.675	2.612	2.492	2.471	2.478	2.497	(41)	
	Ebn at Noon		311	309	306	297	290	285	292	300	301	305	308	310	(42)	
		Edh at Noon	36	33	30	26	24	22	23	27	33	36	37	37	(43)	
(44) All-Sky Solar Radiation	RadAvg		2130	1937	1669	1445	1187	982	1122	1422	1749	1960	2115	2132	(44)	
	RadStd		174	175	101	66	72	72	85	105	143	189	203	212	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.81	N/A	N/A	+1.10	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (28 neighbors)		+0.74	N/A	N/A	N/A	N/A	N/A	N/A	-117	+324	+198

Nomenclature: See separate page